

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/24/2026	DATE SUBMITTED:	09/24/2026
NAME:	Melquisedec D Iquin	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO.	Midterm Skills Exam
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1. PROCEDURE OUTPUT

What procedures did you perform?

Sample output (students are allowed to add more steps as needed.)

- Task 1: Prepare the Server
- Task 2: Create the Prometheus User
- Task 3: Create Prometheus Directories
- Task 4: Download Prometheus
- Task 5: Install the Prometheus Binaries
- Task 6: Install the Prometheus Configuration Files
- Task 7: Validate the Configuration
- Task 8: Create the systemd Service
- Task 9: Start and Enable Prometheus
- Task 10: Verify Prometheus
- Task 11: Access Prometheus Through a Browser

Task 1- 2 - User creation / Update & Upgrade

```

GNU nano 7.2                                prometheus-install.yml
--
- hosts: all
  tasks:
    - name: User creation
      ansible.builtin.user:
        name: iquin
    - name: keys
      authorized_key:
        user: iquin
        state: present
        manage_dir: yes

```

```

i@i iquin-VirtualBox: ~/CPE212_Iquin
i@i iquin-VirtualBox: ~/CPE212_Iquin x i@i iquin-VirtualBox: ~
GNU nano 7.2 prometheus-install.yml
state: present

- name: keys
  authorized_key:
    user: iquin
    state: present
    manage_dir: yes
- name: Upgrade (Ubuntu)
  ansible.builtin.apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == Ubuntu

- name: Update (CentOS)
  dnf:
    name: epel-release
    state: latest
    update_only: yes
  when: ansible_distribution == CentOS

TASK [User creation] *****
[WARNING]: The input password appears not to have been hashed. The 'p
argument must be encrypted for this module to work properly.
ok: [192.168.56.123]
ok: [192.168.56.125]

```

```
iquin@iquin-VirtualBox: ~/CPE212_Iquin
GNU nano 7.2      playbook.yml
---
- name: Install and configure prometheus (Ubuntu & CentOS)
  hosts: monitoring
  become: true
  gather_facts: true

  roles:
    - prometheus
EOF

iquin@iquin-VirtualBox: ~/...
GNU nano 7.2      Debian.yml *
```

```
---
- name: Update apt cache
  ansible.builtin.apt:
    update_cache: true
    cache_valid_time: 3600

- name: Install packages (Ubuntu)
  ansible.builtin.apt:
    name:
      - wget
      - tar
      - curl
      - ufw
    state: present
```

Task 3 - Directories



iquin@iquin-VirtualBox: ~/CPE212_Iquin/roles/prometheus/defaults

iquin@iquin-VirtualBox: ~/CPE212_Iquin/roles...

iquin@iquin-V

GNU nano 7.2

main.yml *

```
---
prometheus_version: "2.53.0"
prometheus_user: "prometheus"
prometheus_group: "prometheus"
prometheus_install_dir: "/opt/prometheus"
prometheus_config_dir: "/etc/prometheus"
prometheus_data_dir: "/var/lib/prometheus"
prometheus_web_listen_port: 9090
prometheus_arch: "linux-amd64"
```

```
- name: Create Directories
  ansible.builtin.file:
    path: "{{item}}"
    state: directory
    owner: "{{prometheus_user}}"
    group: "{{prometheus_group}}"
    mode: "0755"
  loop:
    - "{{prometheus_config_dir}}"
    - "{{prometheus_data_dir}}"
    - "{{prometheus_install_dir}}"

- name: Download Binary tarball
  ansible.builtin.get_url:
    url: "https://github.com/prometheus/prometheus/re>
    dest: "/tmp/prometheus-{{prometheus_version}}.tar>
    mode: "0644"

- name: Extract tarball
  ansible.builtin.unarchive:
    src: "/tmp/prometheus-{{prometheus_version}}.tar.>
    dest: "/tmp/"
```

Tasks 4 & 5 - Downloading and binaries

```
- name: Create Directories
  ansible.builtin.file:
    path: "{{item}}"
    state: directory
    owner: "{{prometheus_user}}"
    group: "{{prometheus_group}}"
    mode: "0755"
  loop:
    - "{{prometheus_config_dir}}"
    - "{{prometheus_data_dir}}"
    - "{{prometheus_install_dir}}"

- name: Download Binary tarball
  ansible.builtin.get_url:
    url: "https://github.com/prometheus/prometheus/re>
    dest: "/tmp/prometheus-{{prometheus_version}}.tar>
    mode: "0644"

- name: Extract tarball
  ansible.builtin.unarchive:
    src: "/tmp/prometheus-{{prometheus_version}}.tar.>
    dest: "/tmp/"
```

```

iquin@iquin-VirtualBox: ~/... x      iquin@iquin-VirtualBox: ~ x v
GNU nano 7.2      main.yml *
- name: Download Binary tarball
  ansible.builtin.get_url:
    url: "https://github.com/prometheus/prometheus/re>
    dest: "/tmp/prometheus-{{prometheus_version}}.tar>
    mode: "0644"

- name: Extract tarball
  ansible.builtin.unarchive:
    src: "/tmp/prometheus-{{prometheus_version}}.tar.>
    dest: "/tmp/"
    remote_src: true
    creates: "/tmp/prometheus-{{prometheus_version}}.>

- name: Copy and promtool binaries
  ansible.builtin.copy:
    src: "/tmp/prometheus-{{prometheus_version}}.{{pr>
    dest: "/usr/local/bin/{{item}}"
    mode: "0755"
    owner: "prometheus_user"
    group: "prometheus_group"
    remote_src: true
  loop:
    - prometheus
    - promtool
-

```

Tasks 6-7

```

iquin@iquin-VirtualBox: ~/... x      iquin@iquin-VirtualBox: ~ x      v
GNU nano 7.2      main.yml *
  group: "prometheus_group"
  remote_src: true
loop:
  - prometheus
  - promtool
- name: Copy console libs
  ansible.builtin.copy:
    src: "tmp/prometheus-{{ prometheus_version }}.{{ >
    dest: "{{ prometheus_config_dir }}/{{ item }}"
    remote_src: true
    owner: "{{prometheus_user}}"
    group: "{{prometheus_group}}"
  loop:
    - consoles
    - console_libraries

- name: Deploy prometheus.yml config
  ansible.builtin.template:
    src: prometheus.yml.j2
    dest: "{{ prometheus_config_dir }}/{{ item }}"
    owner: "{{prometheus_user}}"
    group: "{{prometheus_group}}"
    mode: "0644"
  notify: restart prometheus

- name: Deploy prometheus systemd service unit

```

[^]G Help [^]O Write Out [^]W Where Is [^]K Cut
[^]X Exit [^]R Read File [^]\ Replace [^]U Paste

Task 8 - 9

```

iquin@iquin-VirtualBox: ~/CPE212_Iq...
iquin@iquin-VirtualBox: ~/... x iquin@iquin-VirtualBox: ~ x v
GNU nano 7.2 main.yml *
- console_libraries

- name: Deploy prometheus.yml config
  ansible.builtin.template:
    src: prometheus.yml.j2
    dest: "{{ prometheus_config_dir }}/{{ item }}"
    owner: "{{ prometheus_user }}"
    group: "{{ prometheus_group }}"
    mode: "0644"
  notify: restart prometheus

- name: Deploy prometheus systemd service unit
  ansible.builtin.template:
    src: prometheus.service.j2
    dest: /etc/systemd/system/prometheus.service
    mode: "0644"
  notify:
    - reload systemd
    - restart prometheus

- name: Ensure prometheus service enabled
  ansible.builtin.systemd:
    name: prometheus
    enabled: true
    state: true
    daemon_reload: true

^G Help      ^O Write Out ^W Where Is  ^K Cut
^X Exit      ^R Read File ^\ Replace   ^U Paste
  
```

```
iquin@iquin-VirtualBox: ~/... x      iquin@iquin-VirtualBox: ~ x      v
GNU nano 7.2      main.yml *
- restart prometheus

- name: Ensure prometheus service enabled
  ansible.builtin.systemd:
    name: prometheus
    enabled: true
    state: true
    daemon_reload: true

- name: Open firewall port CentOS (firewalld)
  ansible.posix.firewalld:
    port: "{{prometheus_web_listen_port}}/tcp"
    permanent: true
    state: enabled
    immediate: true
  when: ansible_facts['os_family'] == "RedHat"

- name: Allow port through UFW on Ubuntu
  community.general.ufw:
    rule: allow
    port: "{{prometheus_web_listen_port}}"
    proto: tcp
  when: ansible_facts['os_family'] == "Debian"
  ignore_errors: true
```

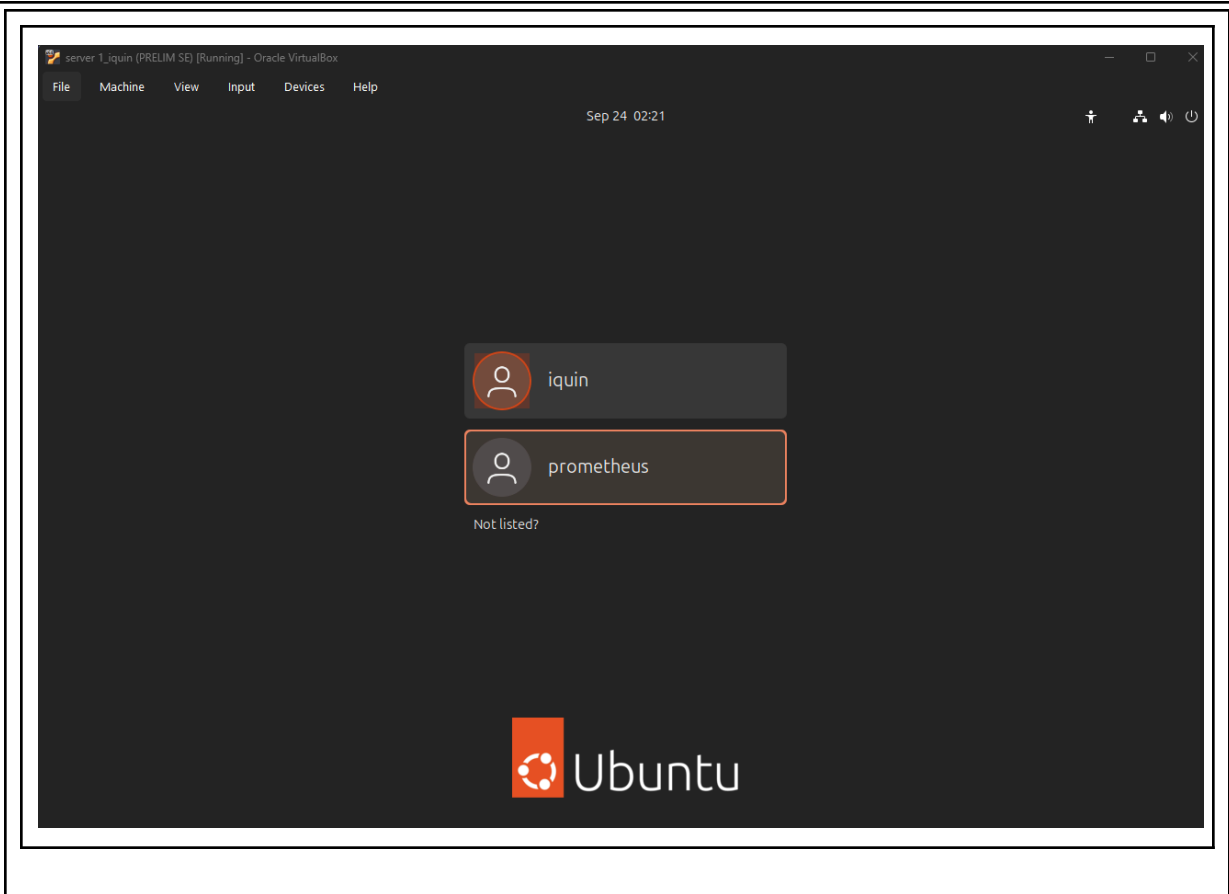
```
iquin@iquin-VirtualBox: ~/... x iquin@iquin-VirtualBox: ~ x
GNU nano 7.2 RedHat.yml *
---
- name: Install package on CentOS
  ansible.builtin.yum:
    name:
      - wget
      - tar
      - curl
      - firewalld
    state: present

- name: firewalld running on CentOS
  ansible.builtin.systemd:
    name: firewalld:
    state: started
    enabled: true
```

Task 10 - 11

```
iquin@iquin-VirtualBox: ~/CPE212_Iq... Q ≡ - □ x
iquin@iquin-VirtualBox: ~/... x iquin@iquin-VirtualBox: ~ x
GNU nano 7.2 main.yml *
---
- name: reload systemd
  ansible.builtin.systemd:
    daemon_reload: true

- name: restart prometheus
  ansible.builtin.systemd:
    name: prometheus
    state: restarted
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

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